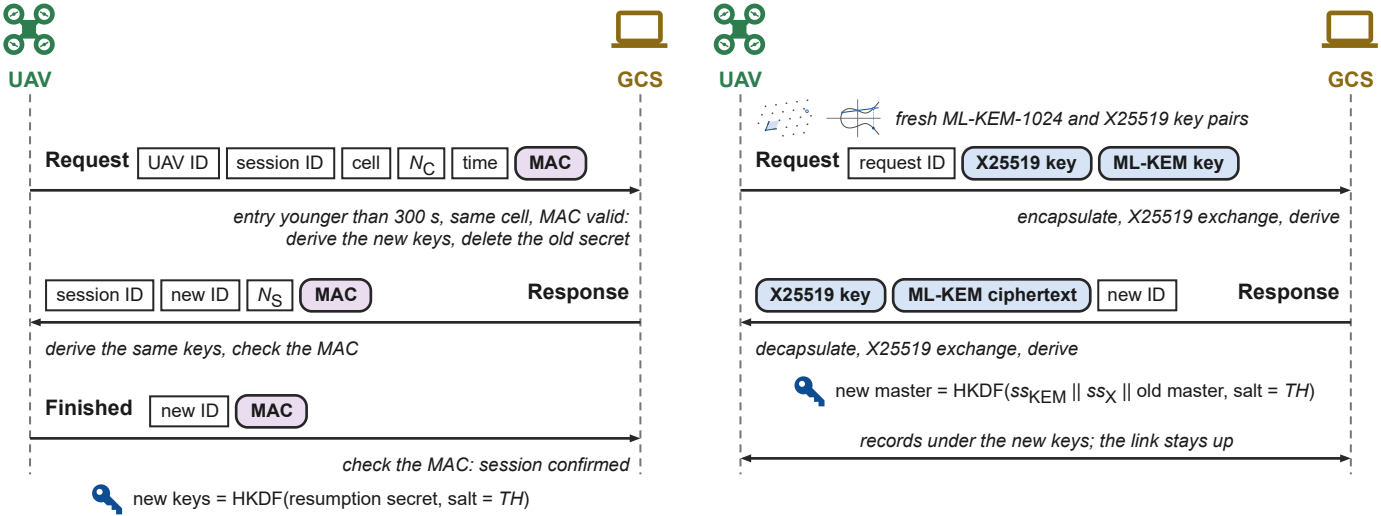



**a** 1-RTT Cached RapidRekey: symmetric only, 324 bytes

**b** PQ ratchet: fresh ML-KEM-1024 + X25519, 4,757 bytes